

Sanath Manjunath

smanj6@uic.edu | 773-718-5083 | LinkedIn: sanath0307 | Portfolio

EDUCATION

University of Illinois Chicago | *MS in Computer Science*

August 2025 – May 2027

- **Relevant Coursework:** Computer Algorithms, Machine Learning, Cloud Computing, Android Development, Adv. SWE Topics

EXPERIENCE

Full Stack Software Engineer

UIC Student Affairs and Technology

Sept 2025 – Present

Chicago, IL

- Architected the SAOnline platform by engineering **REST APIs** and **SQL Stored Procedures**, resulting in a **25%** reduction in data retrieval latency for a more seamless experience.
- Developed and scaled features for the UPASS application using **C#(APS.NET)** and **React**, successfully managing high-concurrency traffic for **65,000+** active users during peak enrollment periods.
- Collaborated with the Data Team to synchronize backend services with student databases.

Full Stack Software Engineer

TopRankers

Aug 2023 – July 2025

Bengaluru, India

- Led end-to-end development of the TopRankers mobile app using **React Native** and **Django**, implementing dynamic data fetching and lazy loading to drive **10,000+** installs and **1,000+** daily active users.
- Revamped the website login and homepage experience and integrated an AI-powered student assistant(chatbot) to help users resolve exam doubts and question queries, to increase engagement.
- Engineered an ML-driven tagging pipeline for **100,000+** questions using **OpenAPI** and automation scripts, reducing manual effort by **40%** improving search relevance and personalized recommendations.
- Optimized platform scalability and achieved a **20% load-time gain** by refactoring **Next.js** modules and offloading heavy workloads to **Celery & Redis** background jobs and tuning **AWS S3** caching.
- Worked closely with cross-functional teams, mentored interns and junior engineers, and presented **technical and business demos** for features shipped to production.

PROJECTS

Production-Grade RAG Pipeline from scratch

Aug 2025 – Nov 2025

SCALA, HADOOP, OLLAMA, APACHE SPARK, APACHE FLINK, NEO4J, AWS (EC2, S3, EMR)

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** platform in **Scala**, integrating Ollama LLMs with FAISS/Lucene for low-latency vector search over research PDFs.
- Engineered scalable data pipelines on **Hadoop** and **AWS EMR** for Map-Reduce based ingestion and ANN index builds, reducing redundant compute with content-hash change detection.
- Implemented **delta-incremental indexing** with content-hash change detection, idempotent upserts, and versioned embedding stores, enabling continuous index refresh while significantly reducing redundancy.

AI Timetable Generator

Jan 2023 – May 2023

- Built an AI-driven scheduling system using **Genetic Algorithms** to auto-generate conflict-free course timetables, implementing constraint satisfaction and optimization heuristics.

SKILLS

Languages: C/C++, Java, Python, JavaScript, TypeScript, Scala, C#, SQL, PHP, Bash

Frameworks: React.js, Next.js, Django, React Native, Node.js, Angular, .NET, Celery, Apache Spark, Flink

Tools & Cloud: AWS (EC2, S3, EMR, Lambda), Azure, GCP, Docker, Kubernetes, Git, CI/CD, Microservices

Data & AI: MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB, Neo4j, Hadoop, Kafka, FAISS